



# Scale Model Parking Garage

## Integrating Automation in Parking Facilities

Bachelor's Thesis

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Subhendu Naskar

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Author       Subhendu Naskar  
Subject      Scale Model Parking Garage  
Supervisor   Mika Oinonen

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## ABSTRACT

Scale model parking garage gives an idea of modern mechanical fundamentals in an educational and research-oriented environment. This thesis aims to develop an automation system of a scale model circular parking garage. The scale model parking garage presents a visual solution to reduce the growing challenges for parking with increasing safety, improving operational efficiency, and enhancing space utilization.

To achieve these aims, the study begins by establishing the fundamental theory of mechanical system, such as ball screw, stepper and servo motors. In addition, it presents the concept of programmable logic controller, along with motion control technologies and human machine interface. Based on these foundations, appropriate mechanical and electrical component are selected, such as AS2021-OD21 servo motor, AM8113-0F20-1002 stepper motor and the Beckhoff CX-series PLC due to its motion control libraries, EtherCAT communication capabilities, and strong real-time performance. Finally, the thesis examines assembling and testing the project.

The results of this work demonstrate the successful designs of the system architecture, development of the control logic and implementation of human-machine interface. The scale model parking garage provides a prototype platform for preparing a foundation for future testing, implementation, expansion and read up on automated parking technologies. Overall, the thesis presents a practical application of automation-based system along with modern parking infrastructure.

Keywords    Scale model parking garage, PLC, I/O systems, servo and stepper motor, ball screw system

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# 1 Introduction

The growing number of vehicles in cities has created formidable problems for parking framework. Traditional parking garages need large areas for circulation track. Due to this drawback, it reduces the efficiency of land use in densely populated metropolitan circumstances. City areas become more expensive and limited. So, there is an increasing demand for creative parking solutions, which can maximize parking capacity while maintaining safety and operational efficiency.

An automated parking system appears as an effective solution to these problems. Unlike old parking garages where drivers manually park their vehicles, an automated parking systems use mechanical equipment as well as control technologies to transport vehicles to their available parking spaces. An automated parking garage can positively increase the parking spaces within a smaller track by terminating internal driving paths and ramps. It also reduces the human error risk and improves safety during vehicle handling.

One example of a well-organized automated parking system is the scale model parking garage. In this layout, vehicles are transported using lifting mechanisms and rotating platforms. Finally, transported vehicles are arranged radially around a central axis. This design enables more efficient use of space compared to the old parking garages. This type of advance parking system is particularly suitable for the city areas where land availability is limited.

The implementation of such a scale model circular parking garage requires the integration of mechanical components with advanced automation technologies. Components like servo motors and ball screw mechanisms provide controlled motion for lifting. Additionally, stepper motors for positioning and rotating the parking platform. A Programmable Logic Controller (PLC) organizes these mechanical movements. It also ensures safe and reliable operation of the system. In this thesis, the scale model circular parking garage is developed with a Beckhoff CX-series PLC. The parking system logic programs are arranged by TwinCAT automation software. Overall, the scale model circular parking garage illustrates PLC-based automation and electromechanical system integration to develop a reliable and efficient automated circular parking garage for educational and research purposes.

## 2 Theoretical foundation

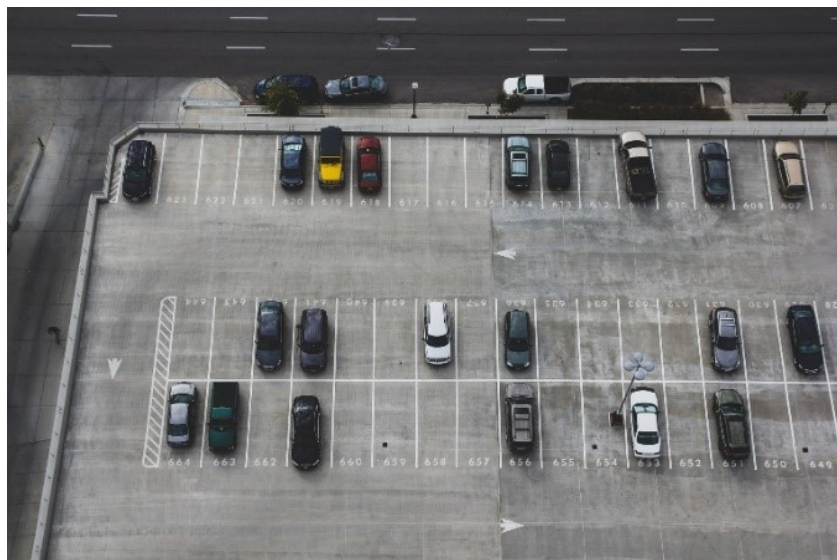
### 2.1 Parking garage

In city areas parking garages are critical infrastructure. They are designed to maximize parking efficiency as well as address the demand for vehicle storage in crowded areas. Day by day, cities expand and traffic increases. So, parking garages have developed to provide both functional and attractive roles in city areas. It helps to reduce street congestion. It also provides efficiency, also more organized, and sustainable representation. (Allik, n.d.)

#### 2.1.1 Traditional parking garages

Traditional parking garages are multi-storey construction, including flat floors, ramps, and space allocated for vehicle parking. Their functioning is dependent on drivers to find the exact location to park vehicles manually. These types of facilities can consume large amount of time as well as occupy huge space due to the need for ramps and circulation paths. (Allik, n.d.)

Figure 1: Traditional Parking Garages. (*Canon EOS 60D + Canon 17-50mm Sample Photo*, n.d.)



### 2.1.2 Automated or mechanized parking garages

Automated parking garages utilize mechanical systems to park and move vehicles, reducing the need for large driving spaces, ramps and aisles within the structure. Automated systems use turntables, conveyors and lifts to position cars, allowing for much denser parking arrangements. A subset of automated garages, leverage a compact, circular parking garages, cylindrical design to further optimize space usage in high-density areas. (Allik, n.d.)

Figure 2: Automated or Mechanized Parking Garages. (*Autostacker's Patented Full-Width, Low-Profile Parking Platform* | Autostacker, n.d.)



### 2.1.3 Circular parking garage

To park the vehicles in a radial arrangement, circular parking garages are designed around a central axis. Using elevators, vehicles are transported vertically and moved into parking based horizontally around the circular structure in such garages.

This layout increases the number of parking spots per square foot and minimizes the space required for ramps. This makes circular parking garages highly efficient in terms of space utilization. (Allik, n.d.)

Figure 3: Circular Parking Garages. (Pansare, 2017)





#### **2.1.4 Mechanical and automation systems in scale model parking garage**

In the scale model parking garage, servo and stepper motors show significant impact in the vehicle transfer mechanism and drive the lift system. The ball screw system provides smooth vehicles positioning and lifting throughout the structure. As the ball screw system operates continuously in high demand conditions, it needs to be reliable and robust all the time during operation. (Ahsan et al., n.d.)

In advanced automated scale model circular parking garages, PLCs are handling the automation part of all vehicle movement. A PLC-based control system supports automated operation with real-time monitoring, as well as, managing all tasks like vehicle rotation, lifting, and parking slot selection. Modern garages need these type of control systems, as they reduce the need for human intervention, decreases manual operations with automated precision. (Supratno et al., 2021)

#### **2.1.5 Benefits of automation and modernization in scale model parking garages**

Scale model circular parking garage can fit in more vehicles within a smaller track due to its automated parking structure, such as a vertical, circular design. This is essential in city areas to decrease the expenses, time and optimize land utilization.

The need for drivers to find an empty parking location and parking in tight spaces, is removed by automation. It also minimizes the risk of car scratches, small damages and accidents due to tight space parking. Moreover, the scale model parking system enhances the overall user experience by adding up the security to prevent unauthorized access of parked vehicles. Scale model parking garages need less lighting, ventilation and heating. So, it becomes a more compact structure, more energy-efficient with fewer ramps. These scale model circular

parking garages also incorporate renewable energy sources, like roof top solar panels to reduce the carbon for sustainable urban development. (GITECH Lifts & Escalators, n.d.)

## 2.2 Mechanical system

The scale model parking garage, mechanical systems, like servo motors, stepper motors, and ball screws, serve a crucial function in positioning and moving vehicles. The integration of mechanical components with automation system is needed for reliable, efficient and safe operation during vehicle movement. In the scale model parking garage, these core components are analyzed and discussed in terms of their roles during the operation. (Budynas et al., 2015)

### 2.2.1 Ball screw systems

In scale model parking garage, ball screw systems are deployed to support in lifting the vehicles with high precision and efficiency. These systems convert rotary motion from the servo motor into linear motion. It enables smooth and controlled movement of elevators. Ball screw system serves several benefits in parking systems. The system offers higher precision, greater load-bearing capacity, and reduced backlash. The motion of ball screw mechanism minimizes friction, improves energy efficiency, and reduces wear. This results in quieter and smoother operation. (Asy et al., 2020)

The parking mechanism ensures reliable and consistent motion by using ball screw systems. It also reduces the risk of jerky movements, ensuring vehicle safety. In addition, these systems requiring less power and distribute force efficiently from the motor while maintaining high levels of control over positioning and speed. (*Ball Screw Linear Stepper Motor Manufacturer - Jkongmotor*, n.d.)

### 2.2.2 Stepper motors

Stepper motors are ideal for incremental and accurate movement applications. In a scale model parking garage, stepper motors are used to control the horizontal movement between the parking spaces. Stepper motors are moved in fixed steps or increments and operated by the pulse from the control system. It ensures smooth position accuracy that is required for parking system application. Stepper motors do not require a feedback system for positioning because these motors can maintain their position based on the pulse count. Stepper motors

provide precise movement control for aligning vehicles in limited spaces. However, stepper motors may not be perfect for applications requiring high torque, and they are typically less powerful than servo motors (Shah, n.d.).

### **2.2.3 Servo motors**

Servo motors are equipped with feedback mechanisms and offer precise control that allows the system to adjust the motor's position, torque and speed by monitoring. In the scale model parking garages, servo motors are specially use for heavy lifting functions, like operating vehicle elevators. Servo motors connected to sensors which provide real-time feedback to the PLC. This precisely adjusting control feedback as well as high torque and durability are essential for high adaptability and accuracy tasks, like levelling a vehicle accurately to align with parking bays. Servo motor responds powerfully to control commands and ensure smooth operation under heavy load. It also contributes efficient and reliable system during operation (Gao et al., 2024).

### **2.2.4 System design and integration of electro-mechanical components**

The mechanical and motor systems in a scale model parking garage are part of a larger electromechanical design. Ball screws, gears, sensors and motors in this system work in conjunction with a control system managed by a PLC. The PLC blends the movements of all components. It also synchronizes actions such as platform rotation, horizontal positioning and vehicle lifting.

The control system must have feedback loops by integrating sensors to monitor the status of each component. For example, encoders provide feedback on the position of the elevator or the platform, ensuring safe vehicle positioning. Control logic is programmed to manage load distribution and prevent collision. It ensures that all components operate within safe parameters. An efficient electromechanical design ensures that motors receive adequate current without overloading the system by considering power distribution. Servo motors require more power during lifting operations. In the scale model parking garage system, safety is important due to heavy mechanical components are taking part in. Manual control options, emergency brakes, and overload protection are integrated to ensure that the system remains safe during a mechanical failure. These safety mechanisms are turned on automatically and operated by the PLC. (Liu & Zhu, 2010; Rouvroye & van den Bliek, 2002)

### 2.2.5 Electro-mechanical system benefits

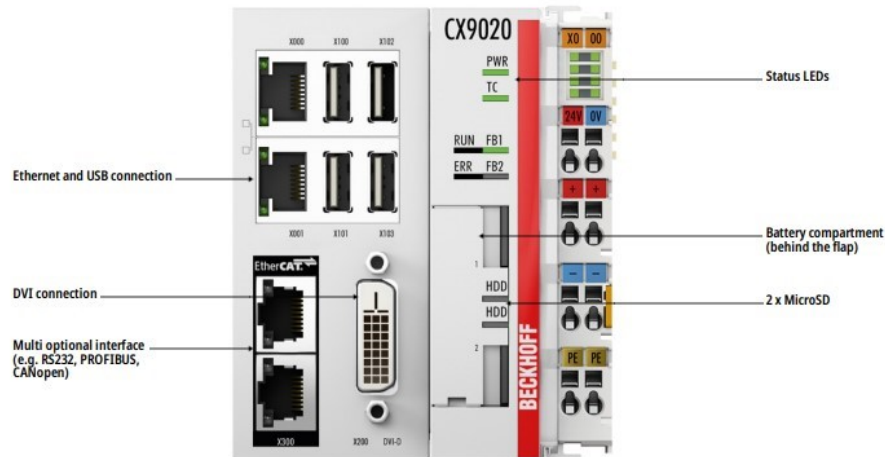
The integration of ball screw servo motors, and stepper motors contribute the essential responsibility for more precise and smoother vehicle handling. This accuracy improves the space used in the scale model parking garage and support to reduce energy consumption. A well-designed electromechanical system can control the high-load conditions. This system builds up with precisely fine-tuned components and high-quality materials to become more resistant to wear. Electromechanical systems with modular designs are adaptable for less routine maintenance, reliability and better enhancement. For instance, motors can be upgraded without a complete system renovate. (Allen, 2005)

## 2.3 Programmable logic controller (PLC)

A Programmable Logic Controller (PLC) is a digital computing system. It is globally used in industrial automation to control processes and machines. PLC provides as the main control unit. In the scale model parking garage, PLC coordinates all automated functions like lifting, positioning and rotating vehicles. Its real-time control capabilities, reliability and flexible programming create the PLC essential for ensuring precise movement, smooth operations, and safety protocols. The PLC connects with mechanical components and sensors to observe the system status continuously. It facilitates error-free and efficient vehicle handling within the parking place and execute instant adjustments.

An industrial computer, PLC is fundamentally invented for controlling field devices and reading data in automated processes system. PLCs vary in size, dependent on the number of input and output (I/O) ports. Larger networked PLCs can support extended racks with hundreds of I/O cards to control more complex operations. On the other hand, smaller PLCs might have a few I/O cards. PLCs are built for harsh industrial environments. PLCs have irregular designs that allow them to go through extreme conditions like dust, moisture, temperature fluctuations, and other challenging factors. PLCs become crucial in high-demand settings such as automated scale model parking garages through their advanced control capabilities and flexibility.

Figure 4: PLC Hardware Components. ([www.duodesign.co.uk](http://www.duodesign.co.uk), n.d.)



### 2.3.1 Beckhoff CX-series PLC

The Beckhoff CX-series PLC was selected for this project based on its suitability for automation and motion control applications. The CX-series PLC integrates the functionality of a traditional PLC with PC-based control features, making it appropriate for complex systems such as scale model parking garages.

The Beckhoff CX-series PLC includes:

- Support for real-time control and deterministic operation
- Compatibility with EtherCAT communication
- Seamless integration with stepper and servo motor systems
- Support for TwinCAT automation software
- Scalability for future system expansion

This PLC is expected to efficiently handle the safety requirements, motion coordination and sequencing of the circular parking system. (Beckhoff Automation GmbH & Co KG, Hülshorstweg 20, 33415 Verl, n.d.-b)

### 2.3.2 TwinCAT automation software

TwinCAT is an automation software, used for PLC programming and system configuration. It provides tools for developing control programs with the help of standardized PLC programming languages like functional block diagram (FBD) and structured text (ST). In this

project, TwinCAT is planned to be used for implementing the safety interlocks, motion sequences and control logic of the parking system. The software includes simulation and debugging, which acknowledges the control program to be tested before physical implementation. This method helps to improve system reliability and reduce programming errors. (Beckhoff Automation GmbH & Co KG, Hülshorstweg 20, 33415 Verl, n.d.-b)

### **2.3.3 Motion control**

Motion control in the proposed scale model parking garage is implemented using a mixture of servo and stepper motor.

The Beckhoff control technology supports motion control through communication interfaces and integrated software functions. It enables synchronized movement between mechanical components and motors. The servo motor needs accurate incremental positioning, but the stepper motor is considered for rotational movement which requires higher torque and continuous feedback. The PLC ensuring smooth motion, accurate positioning, organizes these motors and safe operation of the circular parking structure. (Beckhoff Automation GmbH & Co KG, Hülshorstweg 20, 33415 Verl, n.d.-b)

### **2.3.4 Safety system**

Safety is a vital requirement in circular parking garage systems. The following safety features are required to be implemented through PLC logic:

- Emergency stop functionality
- Limit switch protection to prevent overtravel
- Motor fault detection and system shutdown
- Safe stop behavior during power failure

These safety measures protect vehicles, equipment, and users (Smith & Simpson, 2020).

### **2.3.5 Human-machine interface**

An HMI will be integrated with the PLC to allow user interaction with the parking system. The HMI will provide:

- User commands for parking and retrieval
- Display of available parking slots
- System status information
- Alarm and fault notifications

The PLC will communicate with the HMI to exchange control commands and system feedback (Beckhoff Automation GmbH & Co KG, Hülshorstweg 20, 33415 Verl, n.d.-b).

### **3 Mechanical implementation**

The mechanical system implementation, along with a structured method to ensure authentic system operation and systematic development of the scale model parking garage. The procedure is split up into sequential stages. It begins with a hardware assembly and end with function and validation testing.

In the circular parking garage system, the first stage is mechanical structure assembly. In this case, the installation starts with the mounting of servo, stepper motors and the ball screw mechanism. During the process, ensure the mechanical stability as well as proper alignment to prevent errors. After the mechanical setup, PLC, motors, drives, and the safety devices are connected through electrical cables.

The system undergoes step-by-step testing in the following software development. Individual components, such as PLC and servo motors, are tested independently to verify perfect operation. Afterward, vehicle parking and integrated system testing are performed to evaluate the complete workflow including retrieval sequences. Safety features like emergency stop functionality are tested to confirm proper system response under fault conditions.

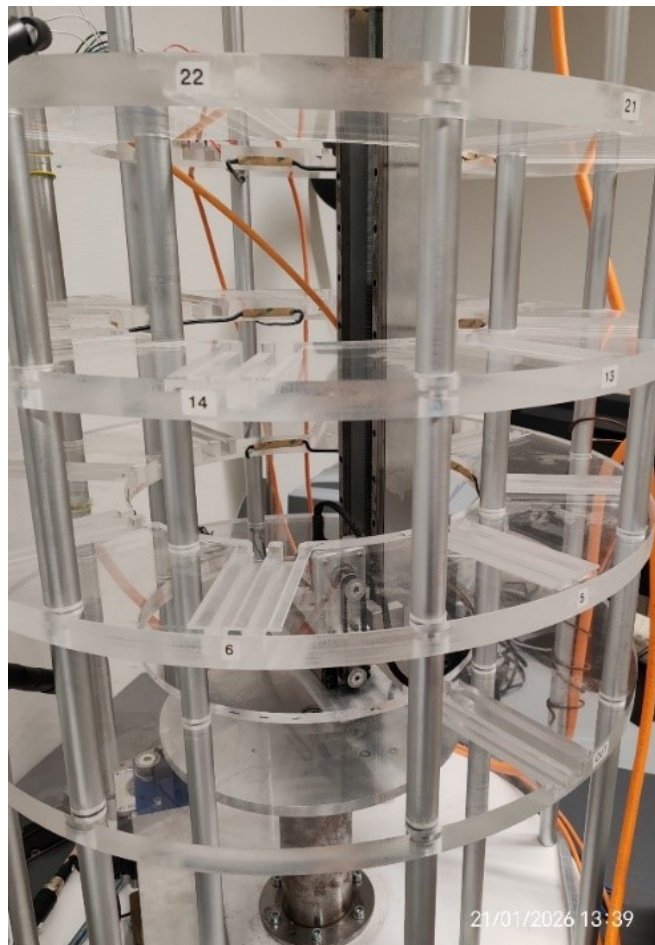
The final stage of the methodology involves improvement and system validation. Mechanical precise tuning and software adjustments help to observe performance limitations or errors. The implemented system runs safely, reliably and in accordance with the objectives defined in the planning phase, verified by this structured workflow.

### 3.1 Mechanical layout

In the scale model circular parking garage, the mechanical layout is designed to perform smooth, precise and stable movement of lift mechanism enclosed by compact structure. The layout is constructed on a cast iron base frame around the circular parking platform to ensure mechanical stability during operation. The design allows sufficient clearance for moving components. It also prioritizes efficient space utilization like the lift mechanism and the rotating platform.

The mechanical structure combination of vertical lifting mechanism, supporting guide rails, and rotating circular platform. These components are organized to minimize mechanical friction during motion. The layout also equips with ball screw assemblies and accessible mounting points for motors and gears. System reliability is maintained through the enhancing of load distribution and component alignment, a process designed to neutralize mechanical tension that could otherwise compromise long-term performance.

Figure 5: Mechanical Parts assembly





### 3.2 Ball screw system assembly and installation

In the scale model parking garage, the ball screw system is implemented as the primary mechanism for vertical motion. Its function is to convert the rotational motion of the driving motor into linear motion. It enables controlled lifting of the parking platform. The ball screw is selected due to its ability to lift axial loads efficiently with low friction and high positioning accuracy.

The ball screw is set up vertically with the help of bearing supports at both ends to reduce vibration and maintain alignment. During assembly the nut is securely attached to the moving platform, and it confirmed that the linear motion is transferred smoothly without backlash. Proper lubrication is applied to ensure the uniform velocity and reduce wear throughout the operating range.

The ball screw is precisely established to prevent binding and irregular loading alignment. This installation ensures precise vertical positioning, lifting mechanism and safe operation.

Figure 6: Ball screw assembly



### 3.3 Stepper motor mechanical integration

Figure 7 demonstrates the use of the stepper motor to rotate the circular parking platform with specific angular control. The motor is durably mounted and responsible for generating rotational motion. These motors ensure stable operation and accurate positioning and during rotation.

On the other hand, figure 8 another stepper motor is mechanically connected to a fork mechanism to control the automated parking door actuation. It is controlled by PLC feedback mechanisms. This linkage authorizes the door to open smoothly and accurately and converts the motor's rotational movement into linear motion.

Altogether, these stepper motor integrations enable safe vehicle movement, correct alignment, ensuring reliable operation and precise control of both door actuation and platform rotation in the scale model automated parking garage system.

Figure 7: Stepper motor assembly for rotation

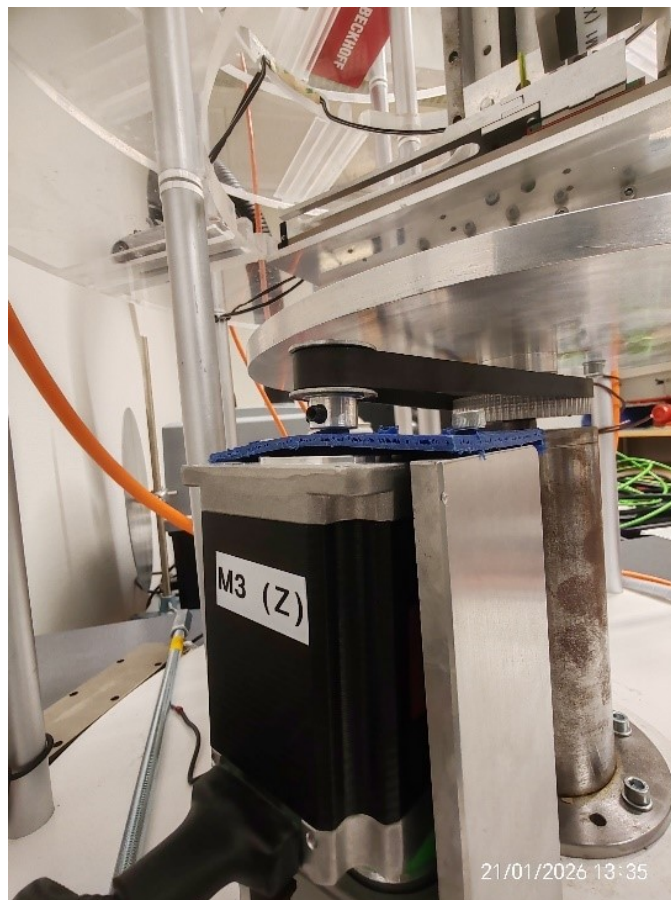
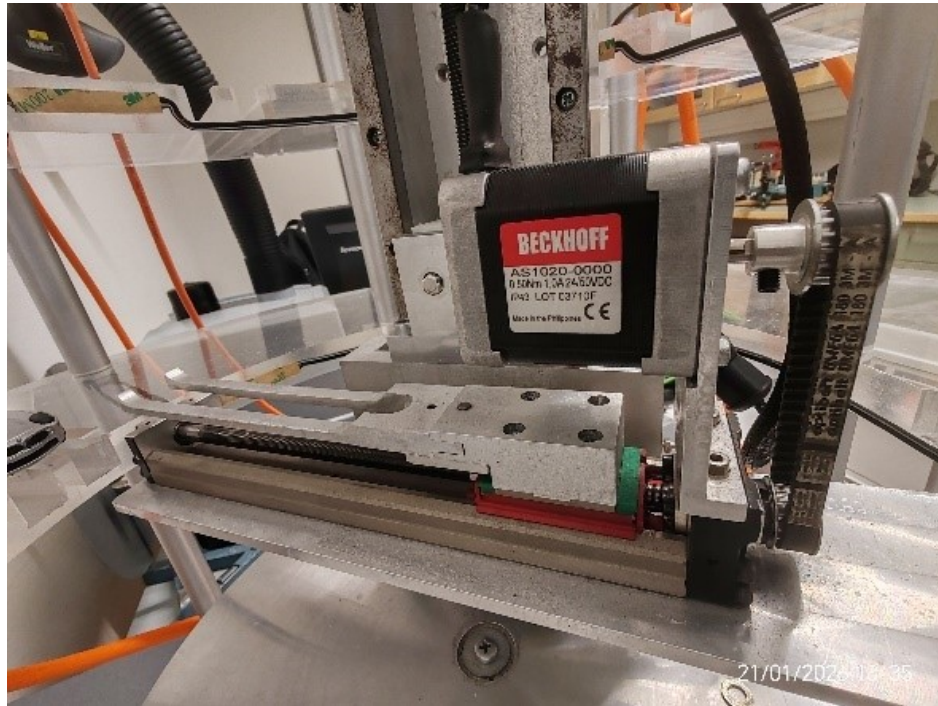


Figure 8: Stepper motor assembly for door open and close



### 3.4 Servo motor mechanical integration

In the scale model parking system, the servo motor is mechanically integrated to perform accurate positioning tasks that require higher torque with fast dynamic response. It is securely mounted on a reinforced bracket. During movement it is designed to withstand operational loads with minimum vibration.

The motion of the servo motor is transmitted to the driven mechanical components through a direct coupling to confirm precise motion control and efficient torque transfer. Proper alignment between the servo motor shaft and the driven mechanism is maintained to avoid wear and reduce mechanical losses caused by misalignment.

This mechanical integration allows smooth motion under varying load conditions. During operation real-time motion and precise positioning correction are achieved due to the communication between servo motor's internal feedback and PLC. As a result, in the scale model automated parking garage system, the servo motor integration enhances overall reliability, safety, and precision.

Figure 9: Servo motor assembly with ball screw system



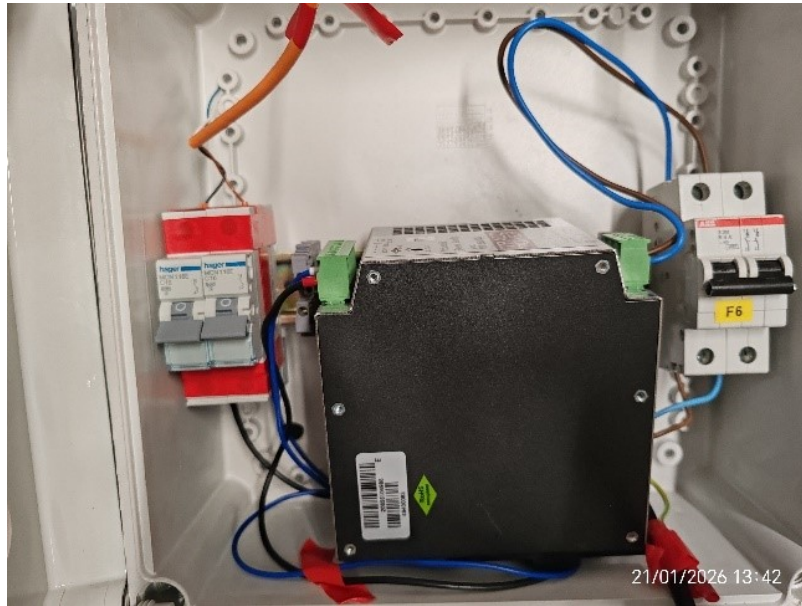
## 4 Electrical and control system Implementation

### 4.1 Power supply selection and distribution

In the scale model circular parking garage, the power supply system is implemented to provide reliable and stable electrical power to all components. A distributed power supply is adopted for the motor drives and control system where different voltage levels are required.

A DC power supply is used to power the control circuits, stepper and servo motor drive, input and output modules and PLC. Power distribution is carried out through fuses and circuit breakers to prevent short circuits and overloads and ensure safe operation. To enhance operator safety and reduce electrical noise proper grounding is implemented throughout the system. According to the figure 10, the power supply layout is assembled to acknowledge easy troubleshooting and future system extension.

Figure 10: SMPS connection diagram



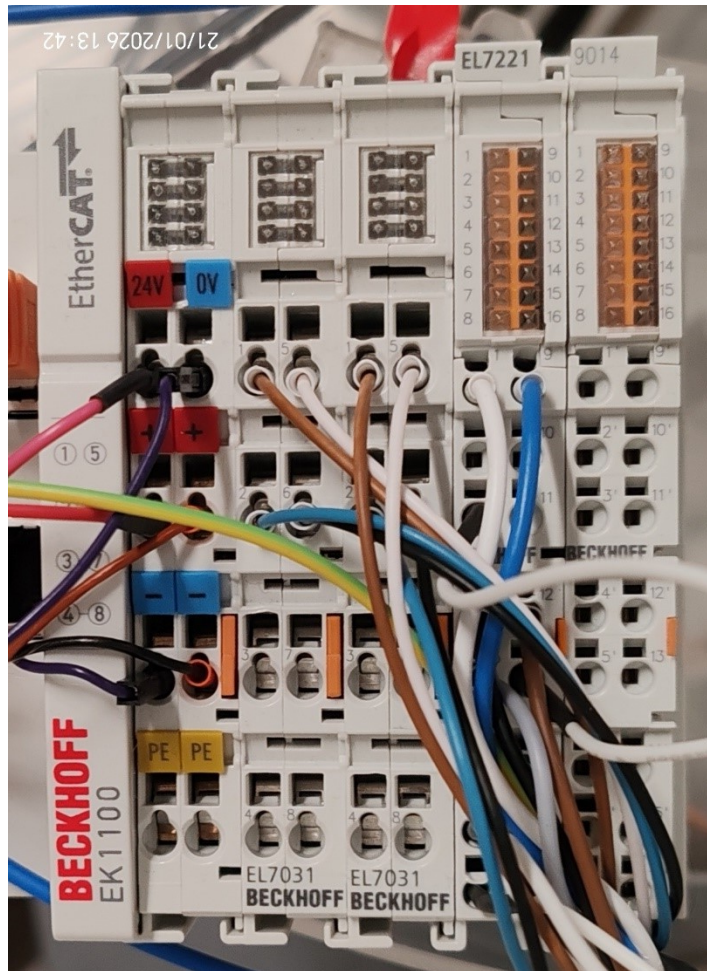
## 4.2 Motor wiring diagram

Both the stepper motor and the servo motor wiring design integrate into a federate control and power distribution framework in the system. A stepper motor operates using an open-loop control system, where electrical pulses are applied in discrete steps to move the motor shaft by fixed increments without feedback. Their wiring is designed to observe for safety, reliability, and signal integrity.

Both motor systems provide as the central communication and signal handling platform with the PLC through beckhoff EtherCAT I/O modules. The EtherCAT network ensures seamless integration of motion commands and feedback signals into the control architecture, real time data exchange, and deterministic timing. The wiring layout also incorporates protective elements to support stable operation, dedicated power supplies, and grounding terminals.

Figure 11: IO connection for stepper and servo motor





## 5 Software implementation

### 5.1 PLC project structure and task configuration

In the scale model circular parking garage system, the Beckhoff TwinCAT 3 software is used to develop logic sequences. The PLC program is organized in a simple and modular structure to make the system easier to understand.

Figure 12 shows, the main Program Organization Units (POUs) structure for the project. Each POU has a specific purpose in the control system:

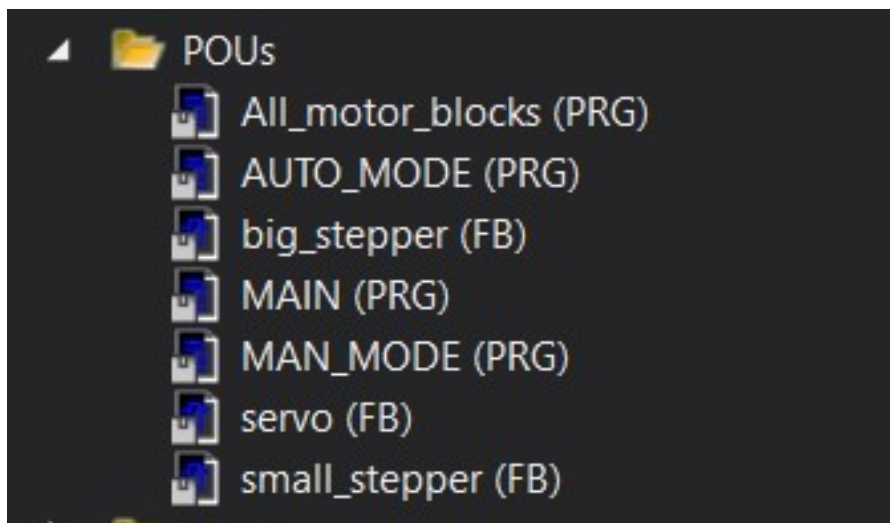
- **MAIN (PRG):**  
Runs the overall system logic and coordinates communication between all program blocks.

- **AUTO\_MODE (PRG):**  
Controls the automatic parking and retrieval sequence.
- **MAN\_MODE (PRG):**  
Allows manual operation for troubleshooting, testing, or calibration.
- **big\_stepper (FB) and small\_stepper (FB):**  
Function blocks that control the stepper motors, including direction and speed monitoring.
- **servo (FB):**  
Controls the servo motor and motion commands and handles encoder feedback.
- **All\_motor\_blocks (PRG):**  
Groups and manages all motor-related function blocks so the system can run them together smoothly.

The PLC program uses a cyclic task that runs the main logic at a fixed interval to ensure reliable and smooth operation. This task handles important actions such as checking safety signals and sending motor commands.

Less critical functions, such as status updates, run in separate tasks so they do not interrupt time sensitive motor control. This structure keeps the system stable, responsive, and easy to extend.

Figure 12: Task configuration in TwinCat Software



## 5.2 Input output configuration

In the system, both input and output configuration is implemented through EtherCAT-based terminals which are connected to the Beckhoff CX series PLC. In the TwinCAT I/O environment, each I/O and control component has a symbolic name. It also allows the software to reference hardware channels in an organized and clear manner.

Digital inputs are used for devices such as emergency stop buttons and entry or exit command buttons. Digital output control elements such as auxiliary actuators, motor enable signals, and indicator lights. Making the program easier to update and maintain and improving readability, symbolic names are used instead of fixed memory addresses.

Figure 13 shows, global variable declarations are used to map key input and control signals. System states and user commands are represented by the variables 'AutoMode', 'ExitBtn', 'EntryBtn', 'ManualMode', and 'ResetButton'. Motor speed parameters 'speedfactor10', 'speedfactor12' and 'speedfactor11' are used for fine-tuning performance. These variables are arranged in the global section to confirm consistent access to the program.

Figure 14 shows, a variable declaration section where five hardware buttons (button\_1 to button\_5) are defined as Boolean inputs and mapped to input addresses using %I\*.

Figure 15 shows, how these button inputs are logically assigned to different control functions, such as reset, manual/auto mode selection, and motor control actions (stop, up, forward, open, down, reverse, and close), by linking each function to a specific button signal

Without requiring major changes to the software design, this structured text input and output address improves debugging efficiency, reduces programming errors and supports future system expansion.

Figure 13: Virtual IO configuration in TwinCat Software



```

1  | attribute 'qualified_only';
2  VAR_GLOBAL
3
4      // Global control
5      ManualMode      : BOOL;
6      ResetButton     : BOOL;    // Reset all stops in manual mode
7      AutoMode        : BOOL;    // Enable auto system
8      ResetBtn        : BOOL;    // Reset all stops in auto mode
9      EntryBtn        : BOOL;
10     ExitBtn         : BOOL;
11     select_slot     : INT;
12     speed_factor_10 : REAL;
13     speed_factor_11 : REAL;
14     speed_factor_12 : REAL;
15 END_VAR
16

```

Figure 14: Hardware IO configuration in TwinCat Software

```

VAR
    // Hardware buttons
    button_1 AT %I* : BOOL;
    button_2 AT %I* : BOOL;
    button_3 AT %I* : BOOL;
    button_4 AT %I* : BOOL;
    button_5 AT %I* : BOOL;
END_VAR

```

Figure 15: Hardware IOs

```

// -----
// link to hardware input
// -----

// data.ResetButton      := button_4;
// data.ManualMode       := button_1;
// data.AutoMode         := NOT(button_1);
// M1_StopButton         := button_3;
// M2_StopButton         := button_3;
// M3_StopButton         := button_3;
// M1_UpButton           := button_5;
// M2_FwdButton          := button_5;
// M3_OpenButton         := button_5;
// M1_DownButton         := button_2;
// M2_RevButton          := button_2;
// M3_CloseButton        := button_2;

```

### 5.3 Motion control

TwinCAT motion control libraries come up with standardized function blocks for controlling stepper and servo motors. These function blocks enable accurate position, speed, control of direction and acceleration. It also maintains synchronization between different motion axes.

Motion sequences are organized by the PLC program like positioning alignment, controlled stop, and platform rotation. After safety conditions are verified motion commands are executed to ensure secure and collision-free operation.

TwinCAT's motion control environment allows parameter tuning and monitoring. It improves the whole systems performance and ensures smooth mechanical movement. This implementation enables precise positioning and continuous motion, which is essential for automated parking operations.

### 5.4 Servo and stepper motor software configuration

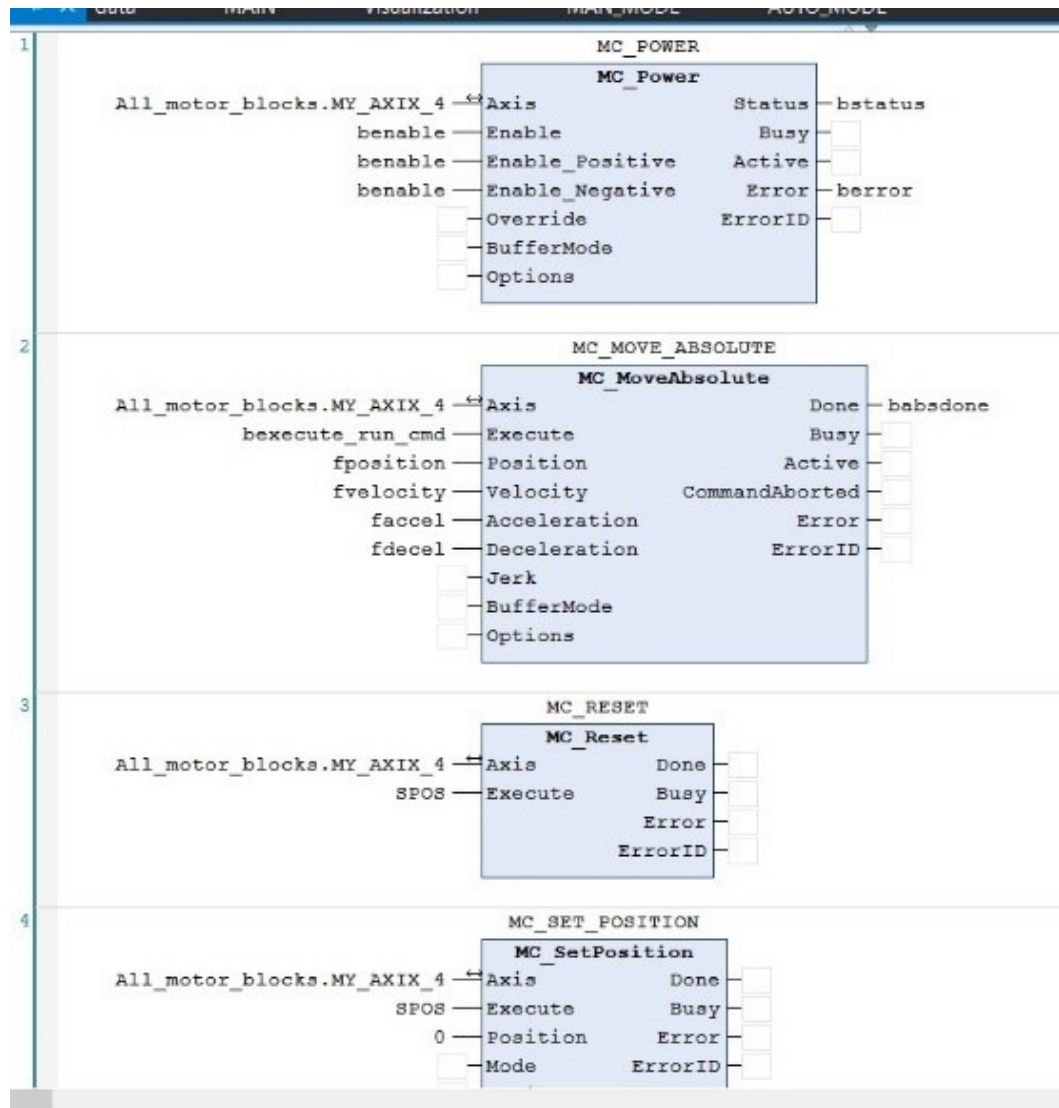
In TwinCAT, the servo motor is configured as a closed-loop motion axis. It connects to the PLC via EtherCAT protocol. As shown in figure 16, standardized PLC motion control function blocks are used to manage axis operation like 'MC\_SetPosition', 'MC\_Reset', 'MC\_MoveAbsolute' and 'MC\_Power' etc. Encoder feedback is continuously processed by the drive to monitor real-time correction of actual position, motion errors, torque and velocity. It also allows accurate trajectory tracking.

Motion parameters such as target position, velocity, acceleration, and deceleration are defined within the PLC logic and moved to the motion function blocks. Based on the parameters the 'MC\_MoveAbsolute' block enables accurate positioning by executing movement command. Conversely, 'MC\_Power' handles axis enabling and operational status monitoring. Additional blocks like 'MC\_Reset' and 'MC\_SetPosition' are implemented to recover faults and to adjust the axis reference position when required.

From the servo drive, fault handling and diagnostic feedback are fully integrated into the PLC program. Status signals like 'Error', 'Done', 'Active', and 'Busy' are monitored to confirm safe execution of motion commands. The control logic automatically disables the axis, prevents unintended movement and protects mechanical components when safety violations or

abnormal conditions take place. This structured software configuration enhances operational safety and high-precision motion control and provides reliability

Figure 16: Servo and stepper motor configuration in TwinCat Software



## 6 HMI implementation

The CP6606 series HMI is implemented to provide a user-friendly interaction layer between the automated parking system and the operator. The HMI allows users to receive feedback on system operation and monitor system status. Integration of the HMI with the PLC improves system control, enhances safety, and operational transparency.

## 6.1 HMI hardware and communication setup

In TwinCAT, the HMI is connected to the Beckhoff CX-series PLC through an industrial Ethernet interface to enable real-time data exchange. The HMI directly allows operator commands and reads and writes PLC variables, such as retrieval, start, stop, and parking commands, to be executed without additional communication programming. System feedback, including slot availability and motor status, are displayed to the user. The communication setup ensures that it includes error-handling mechanisms and low-latency operation to maintain safe system behaviours during connection loss or faults.

## 6.2 System design and navigation

The HMI system design describes clarity, user friendliness, and simplicity. The interface is organized into multiple screens, and each screen shows a specific function. A main overview screen displays the current system's status, including parking slot availability, alarm indicators and motor operation.

Navigation between screens is designed by clearly labelled buttons and using icons. Furthermore, screens provide detailed information on manual and automatic control options. Status indicators are used to quickly observe system conditions like faults, warnings and normal operation. Safety related controls are prominently displayed, such as emergency stop and fault acknowledgements status. The HMI design reduced operator errors and restricted access to critical control functions by guiding users through specific operational sequences.

Overall, the HMI implementation supports safe operation, provides effective monitoring and enhances system usability of the scale model parking system. The human machine interface (HMI) is designed to deliver a natural control environment for structurally and clearly operating the scale model circular parking garage. The interface layout is shaped into well-defined functional sections to support both manual and automatic operation, while maintaining efficient and safe navigation for the user.

The HMI screen is split up into clearly labelled areas that allow operators to quickly understand the selection of operating modes, issue of commands and system status. Visual indicators and numerical displays are used to reduce the probability of operator's error and to ensure simplicity of interaction.

### **6.2.1 Manual mode interface**

The Manual Mode section allows individual mechanical movements and troubleshooting are primarily intended for system setup. This mode provides push buttons controls for:

- Vertical movement
- Horizontal or rotational movement
- Door control

The 'select' option refers to manual commands that are enabled and ensures that only the intended motion is activated. This selection-based viewpoint improves operational safety and reduces accidental operation. An 'emergency stop' button is prominently placed within the manual control area for immediate shutdown of all motions when unsafe conditions occur.

### **6.2.2 Automatic mode interface**

The 'Automatic Mode' section is designed for normal automatic parking and retrieval of the process. The user interacts with the system at a higher level without directly controlling individual motions within this mode. The interface includes:

- A numeric input field for entering the desired parking slot number
- System status display for feedback messages and operation progress
- Entry and Exit command buttons to initiate parking or retrieval sequences

Once a slot number is entered, the PLC executes a predefined control sequence that manages parking slot positioning, rotation and the vehicle movement automatically. Status messages such as 'processing', 'reserved by another user', 'please choose the right parking slot' etc. come up as feedback on system actions.

### **6.2.3 Mode selection and speed control**

A dedicated 'Mode Selection' button allows the operator to switch between 'Automatic' and 'Manual' modes. Visual indicators clearly display the active status to ensure that the operator is always careful of the current system's state. Interlocking logic plays a vital role within the PLC to confirm that only one mode can be active at a time.

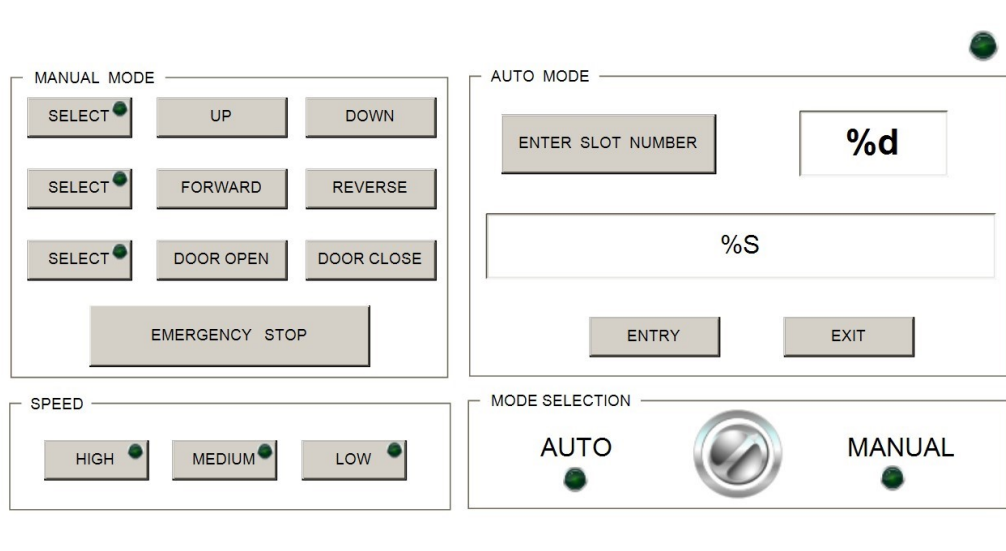
The operator can choose the lift speed such as 'low', 'medium' and 'high' with the help of 'Speed Selection' settings. Depending on system requirements, these speed options allow flexible operation such as higher speed during normal automated operation and slow movement during the testing period. The selected speed modifies the motor parameters through communication to PLC.

#### 6.2.4 Navigation logic and user interaction

HMI navigation logic follows a consistent and simple structure. Operators begin with selecting the operating mode. Buttons are grouped logically to improve system function and usability. It also helps to reduce screen clutter. Display feedback indicators provide real-time confirmation and system responses of user actions. It ensures that operators can easily monitor system behavior and quickly identify the abnormal conditions.

Overall, the HMI system design simplifies the navigations for clarity and prioritizes safety. The specified control sections and well-organized layout support reliable operation of the scale model circular parking garage while facilitating both manual and automated use intervention when required.

Figure 17: HMI design



## **7 Safety system implementation**

Safety is a vital characteristic of scale model parking garage systems during operating of electrical equipment as well as the movement of mechanical components. The safety system implementation ensures reliable operation, prevents potential hazards to users and equipment and protects mechanical components. In this project, safety functions are implemented through a combination of PLC-based software logic and hardware devices.

### **7.1 Emergency stop**

A normally closed emergency stop push button connected to PLC input as an emergency stop function. When the emergency stop button is pressed, it immediately deactivates all drives of the motors and halts system operation.

the PLC program confirms that the motion commands are overridden due to the emergency stop signal indication. The system into a safe state and all motors enable signals are deactivated. The emergency stop status also provides visual feedback to the operator through the HMI.

This implementation ensures quick system shutdown in emergency situations to reduce the risk of mechanical damage or injury.

### **7.2 Motor fault detection**

Motor fault detection is implemented to observe the operational status of both servo and stepper motor systems. Fault signals from motor drives such as overload, communication errors, or overcurrent are communicated to PLC input channels.

The PLC continuously checks the fault signals during system operation. If any fault detects during operation the system disables further motion commands and immediately stops the affected motor. A fault indication is displayed on the HMI to inform the operator of the specific issue.

This fault detection mechanism prevents damages to mechanical components and motors. It also ensures safe system behaviour under fault conditions.

## 8 Testing and validation

System testing and validation are required to recognize the implemented scale model circular parking garage system behaviour under faulty and normal conditions. This phase validates the safety of the mechanical and electrical components as well as the electrical functional performance.

All testing activities are executed in a controlled environment to decrease the risk to equipment and operators. The results obtained from testing are used to assessment system behaviour and verify compliance with the design objectives.

### 8.1 Testing methodology

A structured and step by step testing methodology is implemented to validate system performance. Testing process is initiated with individual component level checking and gradually continuation with full system operation. This method support to prevent damages both mechanical and electrical components along with isolate faults and reduce troubleshooting time.

The testing methodology includes the following steps:

- Component-level testing begins with the verification of individual system components, including motors as well as PLC input and output signals, to ensure their proper functionality.
- Subsystem testing then focuses on evaluating integrated sections of the system, particularly the motion control functions and the operation of HMI, to confirm correct interaction between components.
- Full system testing follows by validating the complete parking and retrieval sequences, ensuring that the entire system operates as intended under normal conditions.
- Finally, safety and fault-condition testing are conducted to verify the system's response to abnormal situations, including the proper functioning of the emergency stop and the appropriate handling of fault conditions.



In TwinCAT software, simulation and online monitoring devices are used throughout the testing process to modify the variable states, observe each task execution and proper communication status during real time data exchange.

## **8.2 Input-output testing**

Both physical and virtual input and output testing is conducted to verify correct signal interchange between the PLC and the field devices. For an example, input device such as an emergency stop button, is manually activated from physical push button as well as from HMI to confirm that the corresponding PLC input signal is received correctly.

Output testing starts with verifying the PLC output signals. Indicators like lamps and motors enable signals, along with direction commands are individually tested to ensure correct operation.

Input and output addressing is cross-checked with the PLC configuration to prevent wiring mismatches. Successful Input and output testing confirms reliable and safe communication between the PLC and field devices.

## **8.3 Motion control testing**

Motion control testing aims on validating the performance of the servo and stepper motor system during the operation. Individual testing is executed without load at low speed to verify correct motor rotation direction, speed control, and positioning accuracy.

In the first case, rotational movement accuracy of a stepper motor is tested by observing position consistency and delivering incremental motion commands through PLC. On the other hand, servo motor control testing initiates by verifying position feedback, deceleration profiles, and smooth acceleration.

During testing, motion parameters such as torque speed, and acceleration limits are adjusted to achieve accurate and stable motion. The PLC's ability to synchronize motor movements is also verified to ensure smooth system operation.

## 8.4 Parking and retrieval sequence testing

Parking and retrieval sequence testing is implemented to confirm the full operational process of the circular parking garage. The verification process covers all major stages of the control logic such as including vehicle entry detection, slot number evaluation, platform rotation direction selection, vertical floor level positioning, and vehicle retrieval execution.

Figure 18 shows, the PLC controller first determines the target floor position based on the selected slot number.

Slots are grouped into ranges. Each range corresponds to predefined vertical target position as already described in the implemented logic. The logic process ensured that the lifting mechanism moved speedily toward the general floor level. After that, it slowed down near the final position to improve accuracy and safety.

Figure 19 shows, the rotation direction of the platform was dynamically selected according to the slot number. Specific slots are activated by reverse rotation or skip rotation command and remaining slots are activated by forward rotation. This type of conditional control behavior improved the system efficiency and decreased the travel distance between slots, along with reduced mechanical wear during both parking and retrieval operations process.

Several test cycles are conducted to evaluate timing synchronization between actuator responses and motion commands feedback. Repeatability tests confirmed that the system consistently reached the correct floor and angular position without deviation. Fault scenarios, such as incomplete motion and incorrect positioning signals are introduced to verify that the controller responded safely and halted the sequence when required.

The successful completion of all tests illustrated that the programmed control logic reliably manages floor positioning, vehicle handling and platform rotation. The results validate that both the parking operation and retrieval process required safety, accuracy, and coordination for continuous automated operation.

Figure 18: Parking floor sequence in TwinCat Software.

```

// Step 40: Move to floor (high speed, then low near)
40:
    // Calculate target based on slot
    IF SlotNumber <= 8 THEN TargetPos := -32000;
    ELSIF SlotNumber <= 16 THEN TargetPos := -64000;
    ELSIF SlotNumber <= 24 THEN TargetPos := -96000;
    ELSE TargetPos := 0;
    END_IF

```

Figure 19: Retrieval sequence in TwinCat Software.

```

// Step 50: Decide rotation direction
50:
    IF (SlotNumber = 1) OR (SlotNumber = 9) OR (SlotNumber = 17) THEN
        Step := 80; // Skip rotation
    ELSIF (SlotNumber = 2) OR (SlotNumber = 3) OR (SlotNumber = 10) OR
        (SlotNumber = 11) OR (SlotNumber = 19) OR (SlotNumber = 18) THEN
        Step := 70; // Reverse rotation
    ELSE
        Step := 60; // Forward rotation
    END_IF

```

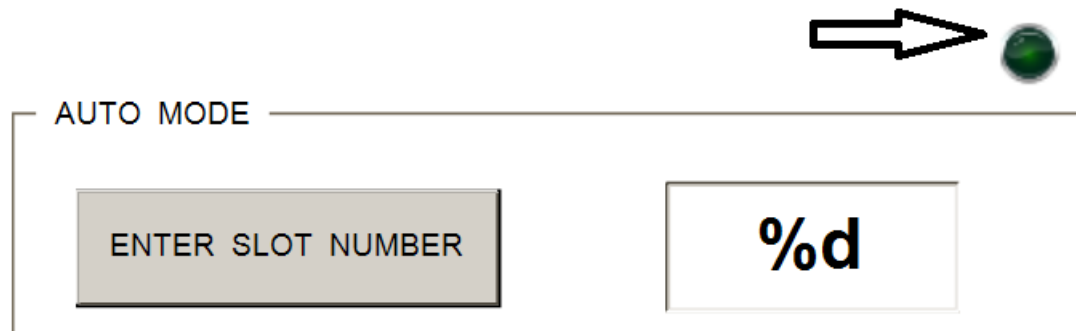
## 8.5 Safety system testing

Safety systems testing is conducted to verify that the protection mechanisms work correctly during both normal operation and fault conditions. When the system is running, the emergency stop circuit is evaluated by activating the emergency push button. The test validated that the PLC immediately disabled motor outputs and stopped platform motion. PLC forced the system into a safe state with visual status indication on the HMI interface.

Within the control system, motor fault detection is also confirmed by simulating motion and drive faults. When a fault condition occurred, the PLC logic successfully identified the abnormal status. Afterwards, the PLC is terminated automatic operation and prevented further movement commands. Figure 20 shows the system status signal, which provides clear feedback regarding the fault condition.

Additional verification confirmed that the whole system restarts only possible after the fault is cleared and safety conditions are restored. The results demonstrate that the implemented safety logic is reliable at shutdown behavior, protects mechanical components, and enhances operator safety during automated parking and retrieval operations.

Figure 20: Green light indicates all system work correctly (NO FAULT).



## 9 Conclusion

Developing this model has been useful, as it demonstrates the benefits of the utilized automation technologies in integrating ball screw mechanisms, servo and stepper motor implementation. It also introduces with Beckhoff CX-series programmable logic controller, including motion control strategy, compact system architecture, and precise control logic development.

Scale model parking garage demonstrates how modern automation technologies improves the safety and efficiency of the automated parking garage. The automation system and the selected hardware components ensure that the scale model parking garage operates safely and move accurately. It also provides real-time monitoring under both manual and automatic mode.

As a prototype, the scale model parking garage demonstrates the effective way for studying and testing of electromechanical parking technologies using a PLC. The whole system has been successfully tested and commissioned during the implementation period. This has provided the author with a good understanding of safety features and a well-designed user interface.

The process of writing and implementing this thesis gave the author valuable knowledge and insights in the field of mechanics and how to integrate stepper and servo motors in different applications. The future, the project can be improved by adding more parking space and more precise control technology along with smart data monitoring.

Finally, the outcomes of this project represent a practical application of a PLC-based prototype automatic parking structure and provide a foundation for further research and development.

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